

GATE DA (Data Science & AI) Practice Question Bank - Set 4

7 Topics · Previous-Year-Style + New Practice Questions · Full Solutions

This set covers four topics: Counting & Basic Probability, Introduction to Machine Learning (Types of Learning & Bias-Variance), the ER Model (entities, attributes, relationships), and Percentages.

SAME HONESTY POLICY AS BEFORE: GATE DA has run only in 2024, 2025 and 2026, and it is a non-coding, MCQ/MSQ/NAT written exam. Every "old" question below is tagged with its real provenance:

[VERIFIED PYQ] -- a genuine, well-documented past GATE question (year and paper stated). Where the source is GATE CSE rather than GATE DA specifically, that is stated explicitly -- these test the identical concept/format used in GATE DA's shared syllabus.

[GATE-style] -- not a verbatim past question (either no verified PYQ exists on that exact sub-point, or the original figure/data could not be reliably reproduced), but written to match real GATE DA difficulty, phrasing, and mark-weightage.

Section B ("New Practice Questions") in every topic is freshly written GATE-level practice -- none of these are claimed to be past questions.

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Topic 1: Counting & Basic Probability

Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE DA 2024, Official Sample Question Paper, Question 7]

Q1. [MCQ · 1 mark]

A fair coin is flipped twice, and it is known that at least one tail is observed. The probability of getting two tails is:

- (A) $1/2$
- (B) $1/3$
- (C) $2/3$
- (D) $1/4$

Answer: (B)

Solution: Sample space = {HH, HT, TH, TT}. Given at least one tail, the reduced sample space is {HT, TH, TT} (3 equally likely outcomes). Of these, only TT has two tails. So $P = 1/3$.

[VERIFIED PYQ — GATE DA 2024]

Q2. [MCQ · 1 mark]

Three fair coins are tossed independently. T is the event that two or more tosses result in heads. S is the event that two or more tosses result in tails. What is $P(T \cap S)$?

- (A) 0
- (B) 0.5
- (C) 0.25
- (D) 1

Answer: (A)

Solution: With only 3 tosses, it is impossible to simultaneously have ≥ 2 heads AND ≥ 2 tails (that would require at least 4 tosses). So $T \cap S = \emptyset$, giving $P(T \cap S) = 0$.

[VERIFIED PYQ — GATE CSE 2003, classic counting question]

Q3. [MCQ · 2 marks]

n couples are invited to a party with the condition that every husband should be accompanied by his wife; however, a wife need not be accompanied by her husband. The number of different gatherings possible at the party is:

- (A) 3^n
- (B) $n \times 2^n$
- (C) $(n+1)^n$
- (D) $2^n \times n!$

Answer: (A)

Solution: For each of the n couples, there are exactly 3 independent possibilities: neither attends, only the wife attends, or both attend (the husband can never attend alone). By the multiplication principle across n independent couples, the total number of gatherings = 3^n .

[GATE-style]

Q4. [NAT · 1 mark]

In how many distinct ways can 5 people be arranged in a row?

Answer: 120

Solution: $5! = 120$.

[GATE-style]

Q5. [NAT · 1 mark]

In how many ways can a committee of 3 be chosen from a group of 8 people (order does not matter)?

Answer: 56

Solution: $C(8,3) = 8!/(3!5!) = 56$.

[GATE-style]

Q6. [MCQ · 2 marks]

The number of distinct ways to arrange the letters of the word "LEVEL" (5 letters, with L repeated twice and E repeated twice) is:

- (A) 30
- (B) 60
- (C) 120
- (D) 20

Answer: (A)

Solution: $5!/(2!2!) = 120/4 = 30$.

[GATE-style]

Q7. [NAT · 1 mark]

A bag contains 4 red balls and 6 blue balls. One ball is drawn at random. Find the probability that it is red.

Answer: 0.4

Solution: $P(\text{red}) = 4/10 = 0.4$.

[GATE-style]

Q8. [NAT · 1 mark]

Two fair dice are rolled. Find the probability that the sum of the two numbers shown is 7 (correct to 4 decimals).

Answer: 0.1667

Solution: Favourable outcomes: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1) = 6. Total outcomes = 36. $P = 6/36 \approx 0.1667$.

[GATE-style]

Q9. [MSQ · 2 marks]

Which of the following are TRUE about basic counting principles?

- (A) The multiplication principle applies when a task consists of a sequence of independent choices/stages
- (B) The addition principle applies when counting the total ways to do one of several mutually exclusive alternatives
- (C) Permutations count arrangements where order matters; combinations count selections where order does not matter
- (D) nP_r is always $\geq {}^nC_r$, for the same n and $r \geq 1$

Answer: (A), (B), (C), (D)

Solution: All four are standard, correct facts. (D) holds since ${}^nP_r = {}^nC_r \times r!$, and $r! \geq 1$ for $r \geq 1$.

[GATE-style]

Q10. [NAT · 1 mark]

A fair coin is tossed 3 times. Find the probability of getting exactly 2 heads.

Answer: 0.375

Solution: $C(3,2) \times (0.5)^3 = 3 \times 0.125 = 0.375$.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

How many distinct 4-digit numbers can be formed using the digits 1,2,3,4,5, without repetition?

Answer: 120

Solution: $P(5,4) = 5 \times 4 \times 3 \times 2 = 120$.

[New Practice]

Q12. [NAT · 2 marks]

How many ways can a 5-card hand be dealt from a standard 52-card deck (order does not matter)?

Answer: 2598960

Solution: $C(52,5) = 52!/(5!47!) = 2,598,960$.

[New Practice]

Q13. [MCQ · 2 marks]

The number of distinct arrangements of the letters of the word "STATISTICS" (10 letters: S×3, T×3, A×1, I×2, C×1) is:

- (A) 50400
- (B) 3628800
- (C) 25200
- (D) 100800

Answer: (A)

Solution: $10!/(3!3!2!1!1!) = 3628800/(6 \times 6 \times 2) = 3628800/72 = 50400$.

[New Practice]

Q14. [NAT · 2 marks]

A box has 3 white and 5 black balls. Two balls are drawn at random without replacement. Find the probability that both are white (correct to 4 decimals).

Answer: 0.1071

Solution: $P = C(3,2)/C(8,2) = 3/28 \approx 0.1071$.

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following are TRUE about the classical (equally-likely-outcomes) definition of probability?

- (A) $P(\text{event}) = (\text{number of favourable outcomes}) / (\text{total number of equally likely outcomes})$
- (B) It requires all outcomes in the sample space to be equally likely
- (C) It can be applied even when outcomes are not equally likely, with no modification needed
- (D) $0 \leq P(\text{event}) \leq 1$, always

Answer: (A), (B), (D)

Solution: (C) is false -- the classical definition is only valid when outcomes ARE equally likely; otherwise a weighted/relative-frequency approach is needed.

[New Practice]

Q16. [NAT · 2 marks]

A card is drawn at random from a standard 52-card deck. Find the probability that it is a king or a heart (correct to 4 decimals).

Answer: 0.3077

Solution: $P(\text{king})=4/52$, $P(\text{heart})=13/52$, $P(\text{king of hearts})=1/52$. $P(\text{king or heart}) = 4/52+13/52-1/52 = 16/52 \approx 0.3077$.

[New Practice]

Q17. [MCQ · 1 mark]

The number of ways to choose a president and a vice-president (two distinct roles) from a group of 10 people is:

- (A) 90
- (B) 45
- (C) 100
- (D) 10

Answer: (A)

Solution: Since the two roles are distinct, order matters: $10 \times 9 = 90$ (a permutation, not a combination).

[New Practice]

Q18. [NAT · 1 mark]

A fair six-sided die is rolled twice. Find the probability that the two rolls show the same number (correct to 4 decimals).

Answer: 0.1667

Solution: Favourable outcomes = 6 (both rolls match), out of 36 total. $P = 6/36 \approx 0.1667$.

[New Practice]

Q19. [MCQ · 2 marks]

In how many ways can 8 distinct books be placed on a shelf such that 2 specific books must always be next to each other?

- (A) 10080
- (B) 5040
- (C) 40320
- (D) 20160

Answer: (A)

Solution: Treat the 2 special books as a single combined unit: 7 units to arrange = $7!$ ways, multiplied by $2!$ for the internal order of the pair. Total = $7! \times 2 = 5040 \times 2 = 10080$.

[New Practice]

Q20. [NAT · 2 marks]

A committee of 4 is to be chosen from 6 men and 4 women. Find the number of ways to choose the committee such that it contains exactly 2 men and 2 women.

Answer: 90

Solution: $C(6,2) \times C(4,2) = 15 \times 6 = 90$.

Topic 2: Intro to ML - Types of Learning & Bias-Variance

Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE DA 2024, Official Sample Question Paper, Question 12]

Q1. [MCQ · 2 marks]

Increasing the regularization coefficient value for a Ridge regressor will: (i) Increase or maintain model bias. (ii) Decrease model bias. (iii) Increase or maintain model variance. (iv) Decrease model variance. Which combination is correct?

- (A) i and iii
- (B) i and iv
- (C) ii and iii
- (D) ii and iv

Answer: (B)

Solution: A larger regularization coefficient pushes the model toward simpler solutions (smaller weight magnitudes). This increases (or at least maintains) bias (i), while making the model less sensitive to the specific training sample, decreasing variance (iv) -- a direct illustration of the bias-variance tradeoff.

[GATE-style]

Q2. [MCQ · 1 mark]

Which of the following best describes SUPERVISED learning?

- (A) Learning from unlabeled data to discover hidden structure
- (B) Learning a mapping from inputs to known (labeled) outputs, using labeled training examples
- (C) Learning purely through trial-and-error interaction with an environment, guided by rewards
- (D) Learning that requires no data at all

Answer: (B)

[GATE-style]

Q3. [MCQ · 1 mark]

Which of the following best describes UNSUPERVISED learning?

- (A) Learning a mapping from labeled input-output pairs
- (B) Finding patterns or structure (e.g. clusters) in data without labeled outputs
- (C) Learning via rewards and penalties received from an environment
- (D) Learning that requires human supervision at every training step

Answer: (B)

[GATE-style]

Q4. [MCQ · 1 mark]

Which of the following best describes REINFORCEMENT learning?

- (A) Learning from a fixed labeled dataset only
- (B) Learning via an agent interacting with an environment, receiving rewards or penalties for its actions
- (C) Learning solely by clustering unlabeled data
- (D) Learning that requires no feedback signal at all

Answer: (B)

[GATE-style]

Q5. [MSQ · 2 marks]

Which of the following are examples of supervised learning tasks?

- (A) Predicting house prices from features (regression)
- (B) Classifying emails as spam or not spam
- (C) Clustering customers into segments based on purchase behaviour, with no labels available
- (D) Predicting whether a tumour is malignant or benign, given a labeled training set

Answer: (A), (B), (D)

Solution: (C) is an unsupervised learning task (clustering with no labels), not supervised.

[GATE-style]

Q6. [MCQ · 1 mark]

The bias-variance tradeoff describes the relationship between:

- (A) Model training time and testing time
- (B) A model's error from overly simplistic assumptions (bias) and its error from sensitivity to the specific training data (variance)
- (C) The number of features and the number of training examples
- (D) The learning rate and the number of iterations

Answer: (B)

[GATE-style]

Q7. [MCQ · 1 mark]

A model with HIGH bias and LOW variance is most likely:

- (A) Overfitting the training data
- (B) Underfitting the training data (too simple to capture the underlying pattern)
- (C) Perfectly fit, with zero error
- (D) Unaffected by the choice of training data, yet still highly accurate

Answer: (B)

[GATE-style]

Q8. [MCQ · 1 mark]

A model with LOW bias and HIGH variance is most likely:

- (A) Underfitting
- (B) Overfitting the training data (very sensitive to the specific training sample)
- (C) The ideal model in all situations
- (D) Immune to changes in the training data

Answer: (B)

[GATE-style]

Q9. [MCQ · 1 mark]

The expected test error (e.g. mean squared error) of a model can be decomposed as:

- (A) $\text{Bias}^2 + \text{Variance} + \text{Irreducible error}$
- (B) $\text{Bias} \times \text{Variance}$
- (C) $\text{Bias} - \text{Variance}$

(D) Only Variance, since bias contributes nothing to test error

Answer: (A)

[GATE-style]

Q10. [MCQ · 2 marks]

As model complexity increases (e.g. fitting a higher-degree polynomial), which of the following is generally TRUE?

- (A) Bias tends to decrease, and variance tends to increase
- (B) Bias tends to increase, and variance tends to decrease
- (C) Both bias and variance always increase together
- (D) Both bias and variance always decrease together

Answer: (A)

Section B — New Practice Questions (10)

[New Practice]

Q11. [MCQ · 1 mark]

Which of the following is an example of unsupervised learning?

- (A) Linear regression on labeled housing-price data
- (B) K-means clustering of customer data with no labels
- (C) Logistic regression for spam classification using labeled emails
- (D) Training a classifier on labeled images of cats and dogs

Answer: (B)

[New Practice]

Q12. [MCQ · 1 mark]

In reinforcement learning, the signal that guides the agent's learning is called:

- (A) The label
- (B) The reward (or reward signal)
- (C) The cluster centroid
- (D) The regularization term

Answer: (B)

[New Practice]

Q13. [MSQ · 2 marks]

Which of the following statements about overfitting are TRUE?

- (A) An overfit model performs very well on training data but poorly on unseen test data
- (B) Overfitting is associated with high variance
- (C) Overfitting is associated with low variance
- (D) Regularization is a common technique used to reduce overfitting

Answer: (A), (B), (D)

[New Practice]

Q14. [MCQ · 1 mark]

Cross-validation is primarily used to:

- (A) Artificially increase the size of the training dataset
- (B) Get a more reliable estimate of a model's generalization performance by testing on multiple held-out subsets of the data
- (C) Eliminate the need for a test set entirely and permanently
- (D) Guarantee zero bias in the resulting model

Answer: (B)

[New Practice]

Q15. [MCQ · 1 mark]

Which of the following is generally considered a sign of UNDERFITTING?

- (A) Low training error and low test error
- (B) High training error AND high test error (the model is too simple to even fit the training data well)
- (C) Low training error but high test error
- (D) High training error but low test error

Answer: (B)

[New Practice]

Q16. [MCQ · 1 mark]

Which of the following is generally considered a sign of OVERFITTING?

- (A) Low training error and low test error
- (B) High training error and high test error
- (C) Low training error but noticeably higher test error (the model has memorized the training data)
- (D) High training error but low test error

Answer: (C)

[New Practice]

Q17. [MSQ · 2 marks]

Which of the following techniques can help reduce overfitting?

- (A) Adding regularization (e.g. L1/L2 penalty)
- (B) Increasing the amount of training data
- (C) Increasing model complexity indefinitely
- (D) Using cross-validation to select hyperparameters

Answer: (A), (B), (D)

Solution: (C) would generally worsen overfitting, not reduce it.

[New Practice]

Q18. [MCQ · 1 mark]

A "semi-supervised" learning setting refers to:

- (A) Training using only labeled data
- (B) Training using only unlabeled data
- (C) Training using a combination of a small amount of labeled data and a larger amount of unlabeled data
- (D) Training with no data at all

Answer: (C)

[New Practice]

Q19. [MCQ · 1 mark]

The "no free lunch" intuition in machine learning broadly suggests that:

- (A) A single algorithm is guaranteed to be the best choice for every possible problem/dataset
- (B) No single learning algorithm is universally superior across all possible tasks/datasets; performance depends on the problem
- (C) Free/open-source ML libraries always outperform paid ones
- (D) All models have exactly the same bias and variance

Answer: (B)

[New Practice]

Q20. [MCQ · 2 marks]

If a model has very high bias (is underfitting), which of the following is a reasonable remedy?

- (A) Increase model complexity (e.g. add more features/higher-order terms), or reduce regularization strength
- (B) Reduce model complexity further
- (C) Collect less training data
- (D) Increase regularization strength further

Answer: (A)

Topic 3: ER Model – Entities, Attributes & Relationships

Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE CSE 2012]

Q1. [MCQ · 1 mark]

Which of the following is INCORRECT about an ER Model?

- (A) An attribute of an entity can have more than one value
- (B) An attribute of an entity can be composite
- (C) In a row of a relational table, an attribute can have more than one value
- (D) In a row of a relational table, an attribute can have exactly one value, or NULL

Answer: (C)

Solution: Statement (C) is the INCORRECT one -- in the relational model, every attribute value in a row must be atomic (a single value, or NULL); this is precisely the requirement of First Normal Form. Multi-valued and composite attributes are allowed conceptually in the ER model (A, B are correct statements about the ER model), but once mapped to the relational model, values must be atomic (D is also a correct statement).

[VERIFIED PYQ — GATE CSE 2004, classic ERD-to-relational mapping question]

Q2. [NAT · 2 marks]

An ER diagram has entities E1 and E2 related by a relationship R of cardinality 1:m. E1's attributes are A11 (key), A12, A13. E2's attributes are A21 (key), A22, and A23 (multi-valued). R itself has no attributes. What is the minimum number of tables needed to represent this schema relationally, with every table satisfying 3NF?

Answer: 3

Solution: Table 1: E1(A11,A12,A13). Table 2: E2(A21,A22,A11) -- since R is 1:m with no attributes, its foreign key (A11) is simply absorbed into the "many" side E2, avoiding a separate table for R. Table 3: the multi-valued attribute A23 must be placed in its own table (with A21 as a foreign key), since a single relational attribute cannot hold multiple values while remaining in 3NF. Total = 3 tables.

[VERIFIED PYQ — GATE CSE 2020]

Q3. [MCQ · 1 mark]

Which of the following is used to represent the supporting (identifying) many-to-one relationship of a weak entity set in an entity-relationship diagram?

- (A) Diamonds with double/bold border
- (B) Rectangles with double/bold border
- (C) Ovals with double/bold border
- (D) Ovals that contain underlined identifiers

Answer: (A)

Solution: A weak entity set itself is drawn as a rectangle with a double/bold border, but the identifying relationship connecting it to its owner (strong) entity set is drawn as a diamond with a double/bold border.

[VERIFIED PYQ — GATE CSE 2018]

Q4. [MCQ · 2 marks]

In an ER model, suppose R is a many-to-one relationship from entity set E1 to entity set E2. Assume E1 and E2 participate totally in R, and the cardinality of E1 is greater than the cardinality of E2.

Which of the following is TRUE about R?

- (A) Every entity in E1 is associated with exactly one entity in E2
- (B) Some entity in E1 is associated with more than one entity in E2
- (C) Every entity in E2 is associated with exactly one entity in E1
- (D) Some entity in E2 is associated with more than one entity in E1, but not every entity

Answer: (A)

Solution: Since R is many-to-one from E1 to E2, each entity in E1 maps to at most one entity in E2. Combined with E1's total participation in R, every entity in E1 is associated with EXACTLY one entity in E2.

[GATE-style]

Q5. [MCQ · 1 mark]

In the ER model, an "entity" represents:

- (A) A relationship between two tables
- (B) A real-world object or concept that can be distinctly identified
- (C) A single attribute value
- (D) A database query

Answer: (B)

[GATE-style]

Q6. [MCQ · 1 mark]

A "composite attribute" in the ER model is an attribute that:

- (A) Can take multiple values simultaneously for a single entity
- (B) Can be broken down into smaller sub-attributes (e.g. "Address" into Street, City, ZIP)
- (C) Is derived from other attributes
- (D) Cannot be stored in the database

Answer: (B)

[GATE-style]

Q7. [MCQ · 1 mark]

A "derived attribute" in the ER model is an attribute whose value:

- (A) Is entered manually by the user only
- (B) Can be computed/derived from other attributes or entities (e.g. Age derived from Date of Birth)
- (C) Is always NULL
- (D) Must be a key attribute

Answer: (B)

[GATE-style]

Q8. [MCQ · 1 mark]

A "multi-valued attribute" in the ER model is represented in an ER diagram using:

- (A) A single-line oval
- (B) A double-line (double-bordered) oval
- (C) A rectangle
- (D) A diamond

Answer: (B)

[GATE-style]

Q9. [MSQ · 2 marks]

Which of the following are TRUE about weak entity sets in the ER model?

- (A) A weak entity set does not have a sufficient set of attributes to form its own primary key
- (B) A weak entity set depends on a strong (owner) entity set for its identification, via an identifying relationship
- (C) A weak entity set's discriminator (partial key), combined with the owner entity's primary key, forms the weak entity's complete key
- (D) A weak entity set can exist and be identified independently of any other entity set

Answer: (A), (B), (C)

Solution: (D) is false -- being identifiable independently of any other entity set is precisely the definition of a STRONG entity set, not a weak one.

[GATE-style]

Q10. [MCQ · 1 mark]

In a relationship set, the "degree" of the relationship refers to:

- (A) The number of attributes in the relationship
- (B) The number of entity sets participating in the relationship (e.g. binary=2, ternary=3)
- (C) The cardinality ratio (1:1, 1:m, m:n)
- (D) The number of instances of the relationship

Answer: (B)

Section B — New Practice Questions (10)

[New Practice]

Q11. [MCQ · 1 mark]

A relationship connecting three different entity sets (e.g. Supplier, Part, Project) is called a:

- (A) Binary relationship
- (B) Unary (recursive) relationship
- (C) Ternary relationship
- (D) Weak relationship

Answer: (C)

[New Practice]

Q12. [MCQ · 1 mark]

A relationship where an entity set is related to itself (e.g. "Employee supervises Employee") is called a:

- (A) Binary relationship
- (B) Unary (recursive) relationship
- (C) Ternary relationship
- (D) Composite relationship

Answer: (B)

[New Practice]

Q13. [MSQ · 2 marks]

Which of the following are valid types of cardinality ratios (mapping cardinalities) for a binary relationship?

- (A) One-to-one (1:1)
- (B) One-to-many (1:m)
- (C) Many-to-many (m:n)
- (D) Zero-to-zero (0:0)

Answer: (A), (B), (C)

Solution: (D) "Zero-to-zero" is not a standard cardinality ratio type.

[New Practice]

Q14. [MCQ · 1 mark]

"Total participation" of an entity set E in a relationship R means:

- (A) Every entity in E must participate in at least one relationship instance of R
- (B) At least one entity in E participates in R, but not necessarily all
- (C) No entity in E participates in R
- (D) E has no attributes

Answer: (A)

[New Practice]

Q15. [MCQ · 1 mark]

"Partial participation" of an entity set E in a relationship R means:

- (A) Every entity in E must participate in R
- (B) Only some entities in E need to participate in R; participation is optional for the others
- (C) E cannot participate in R at all
- (D) R has no cardinality constraint

Answer: (B)

[New Practice]

Q16. [NAT · 1 mark]

An ER diagram has 3 strong entity sets and 2 weak entity sets. If each entity set (strong or weak) maps to exactly one table in the relational schema (ignoring relationship tables for this count), how many entity tables are needed in total?

Answer: 5

Solution: 3 (strong) + 2 (weak) = 5 entity tables.

[New Practice]

Q17. [MCQ · 1 mark]

In an ER diagram, a key attribute is typically represented by:

- (A) A dashed oval
- (B) An underlined attribute name
- (C) A double rectangle
- (D) A shaded diamond

Answer: (B)

[New Practice]

Q18. [MCQ · 2 marks]

Which of the following about ternary relationships is TRUE?

- (A) A ternary relationship can always be losslessly decomposed into three separate binary relationships
- (B) A ternary relationship connects exactly three entity sets and, in general, cannot always be decomposed into binary relationships without losing information
- (C) A ternary relationship is the same as three separate 1:1 relationships
- (D) Ternary relationships are not allowed in the ER model

Answer: (B)

[New Practice]

Q19. [MCQ · 2 marks]

An "aggregation" construct in the ER model is used to:

- (A) Delete redundant entities
- (B) Treat a relationship set (along with its participating entity sets) as a higher-level abstract entity, allowing it to participate in further relationships
- (C) Merge two entity sets into a single entity set
- (D) Represent a many-to-many relationship only

Answer: (B)

[New Practice]

Q20. [MCQ · 1 mark]

Which of the following is generally NOT a valid category of attribute in the ER model?

- (A) Simple attribute
- (B) Composite attribute
- (C) Recursive attribute
- (D) Derived attribute

Answer: (C)

Solution: Simple, composite, derived, single-valued, and multi-valued are all standard ER attribute categories; "recursive" instead describes a type of RELATIONSHIP (an entity set related to itself), not a category of attribute.

Topic 4: Percentages

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [NAT · 1 mark]

What is 25% of 480?

Answer: 120

Solution: $0.25 \times 480 = 120$.

[GATE-style]

Q2. [NAT · 1 mark]

If 40% of a number is 96, find the number.

Answer: 240

Solution: $0.4x = 96 \Rightarrow x = 96/0.4 = 240$.

[GATE-style]

Q3. [MCQ · 2 marks]

A number is increased by 20% and then decreased by 20%. Find the net percentage change.

- (A) 0%
- (B) -4%
- (C) +4%
- (D) -20%

Answer: (B)

Solution: Let $x=100$. After +20%: 120. After -20% of 120 (=24): $120-24=96$. Net change = $96-100 = -4\%$.

[GATE-style]

Q4. [NAT · 1 mark]

In an election between two candidates, one candidate got 60% of the total valid votes. If 8000 valid votes were cast in total, find the number of votes the winning candidate received.

Answer: 4800

Solution: $0.6 \times 8000 = 4800$.

[GATE-style]

Q5. [MCQ · 2 marks]

If A's income is 25% more than B's income, then B's income is what percent less than A's?

- (A) 20%
- (B) 25%
- (C) 30%
- (D) 15%

Answer: (A)

Solution: Let $B=100$, so $A=125$. B is less than A by 25, which as a percentage of A is $25/125 \times 100 = 20\%$.

[GATE-style]

Q6. [NAT · 2 marks]

The population of a town increases by 10% each year. If the current population is 5000, find the population after 2 years.

Answer: 6050

Solution: $5000 \times 1.1 \times 1.1 = 5000 \times 1.21 = 6050$.

[GATE-style]

Q7. [MCQ · 2 marks]

A student needs 40% of the total marks to pass. He scored 130 marks and failed by 30 marks. Find the total marks.

- (A) 400
- (B) 350
- (C) 450
- (D) 500

Answer: (A)

Solution: Passing marks = $130 + 30 = 160 = 40\%$ of total $\Rightarrow$ total = $160 / 0.4 = 400$.

[GATE-style]

Q8. [MSQ · 2 marks]

Which of the following statements about percentages are TRUE?

- (A) Percentage means "per hundred"
- (B) A 50% increase followed by a 50% decrease always returns the value to its original amount
- (C) $x\%$ of y equals $y\%$ of x
- (D) To convert a fraction to a percentage, multiply it by 100

Answer: (A), (C), (D)

Solution: (B) is false -- e.g. 100 increased by 50% = 150, then decreased by 50% of 150 (=75) gives 75, not 100; successive percentage changes are not simply reversible by their negation.

[GATE-style]

Q9. [NAT · 2 marks]

A shirt's price is first increased by 25% and then decreased by 20%. Find the net percentage change.

Answer: 0

Solution: Let $x=100$. After +25%: 125. After -20% of 125 (=25): $125 - 25 = 100$. Net change = 0%.

[GATE-style]

Q10. [NAT · 1 mark]

15% of 200 is what percent of 60?

Answer: 50

Solution: 15% of 200 = 30. As a percentage of 60: $30/60 \times 100 = 50\%$.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

What is 35% of 600?

Answer: 210

Solution: $0.35 \times 600 = 210$.

[New Practice]

Q12. [NAT · 2 marks]

If 20% of a number is 50, find 50% of the same number.

Answer: 125

Solution: $0.2x = 50 \Rightarrow x = 250$. 50% of 250 = 125.

[New Practice]

Q13. [MCQ · 2 marks]

The price of an item is decreased by 10% and then increased by 10%. Find the net percentage change.

- (A) 0%
- (B) -1%
- (C) $+1\%$
- (D) -10%

Answer: (B)

Solution: Let $x = 100$. After -10% : 90. After $+10\%$ of 90 ($=9$): $90 + 9 = 99$. Net change = $99 - 100 = -1\%$.

[New Practice]

Q14. [NAT · 1 mark]

In an exam, a student scores 72 out of 90. Find the percentage score.

Answer: 80

Solution: $72/90 \times 100 = 80\%$.

[New Practice]

Q15. [MCQ · 2 marks]

If B's salary is 20% less than A's salary, then A's salary is what percent more than B's?

- (A) 20%
- (B) 25%
- (C) 30%
- (D) 15%

Answer: (B)

Solution: Let $A = 100$, so $B = 80$. A exceeds B by 20, which as a percentage of B is $20/80 \times 100 = 25\%$.

[New Practice]

Q16. [NAT · 1 mark]

A number increased by 15% becomes 230. Find the original number.

Answer: 200

Solution: $1.15x = 230 \Rightarrow x = 230/1.15 = 200$.

[New Practice]

Q17. [MCQ · 1 mark]

In a class of 50 students, 60% are boys. Find the number of girls.

- (A) 20
- (B) 25
- (C) 30
- (D) 15

Answer: (A)

Solution: Boys = $0.6 \times 50 = 30$. Girls = $50 - 30 = 20$.

[New Practice]

Q18. [MSQ · 2 marks]

Which of the following are TRUE about percentage-change calculations?

- (A) Percentage increase = $(\text{Increase} / \text{Original value}) \times 100$
- (B) Percentage decrease = $(\text{Decrease} / \text{Original value}) \times 100$
- (C) Percentage change should always be calculated based on the NEW value, not the original value
- (D) A percentage change can be negative, indicating a decrease

Answer: (A), (B), (D)

Solution: (C) is false -- percentage change is conventionally calculated relative to the ORIGINAL (starting) value, not the new value.

[New Practice]

Q19. [NAT · 2 marks]

A's salary is reduced by 25%. By what percentage must it now be increased to restore it to the original value (correct to 2 decimals)?

Answer: 33.33

Solution: Let original = 100. After the reduction: 75. To go from 75 back to 100, the required increase is $25/75 \times 100 = 33.33\%$.

[New Practice]

Q20. [NAT · 1 mark]

45 is what percentage of 180?

Answer: 25

Solution: $45/180 \times 100 = 25\%$.
